

Ezra Apple

Software Engineer | AI Product & Agent Infrastructure

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EDUCATION

University of California, Berkeley

May 2025

B.S. in Electrical Engineering and Computer Science

EXPERIENCE

Repl0 (YC S21) | *Software Engineer, AI Agent Platform*

Oct 2025 – Present

- Built and launched **storefront creation and editing workflows** for Repl0's customer-facing AI website builder, including Component Library, reusable templates, and live page previews.
- Designed **persistent user and project memory** for Repl0 Agent, combining scoped context, automatic curation, and on-demand retrieval to retain knowledge across sessions.
- Led consolidation of Repl0's **internal agent skills and development workflows**, making engineering, design, and support knowledge discoverable and reusable across coding agents.
- Improved **preview and publishing reliability** for Repl0's AI website builder, building automatic preview recovery, asset-readiness checks, and observability for diagnosing customer failures.

Decyphr AI | *Co-Founder & CTO*

Jan 2025 – Aug 2025

- Co-founded AI video translation startup, accepted into **UC Berkeley SkyDeck Pad-13** incubator program.
- Owned end-to-end MVP for AI-driven video translation across **25+ languages**, spanning authentication, media upload, processing orchestration, and launch site.
- Architected a serverless media-processing pipeline on **AWS (S3, Lambda, Step Functions, SES)** with fault-tolerant retries for long-running translation jobs.
- Ran **closed beta with 10+ content creators**, translating feedback into product refinements and go-to-market validation.

Lawrence Berkeley National Laboratory | *Machine Learning Researcher*

Oct 2023 – Jun 2025

- Applied **random forests and Bayesian optimization** to ion-source diagnostic data, modeling accelerator performance and exploring configurations to maximize ion output.
- Presented findings to senior researchers supporting **heavy-element synthesis research**, including the element 120 program.

PROJECTS

ShoutOut | *Local-First macOS Dictation App*

- Built and shipped a local-first macOS dictation product in **Swift** with global Fn/Globe capture, AVAudioEngine recording, focused-app paste, and WPM/latency stats.
- Designed a low-friction interaction model with hold-to-record, double-tap hands-free mode, menu bar controls, recording overlay feedback, and first-run permission onboarding.
- Implemented swappable on-device transcription engines using **Apple Speech**, **Apple Dictation**, and **WhisperKit/Core ML**, with dictionary-aware cleanup and spoken-punctuation formatting.
- Managed macOS releases with signed and notarized builds and automatic updates through **Sparkle**.

SKILLS

Languages: TypeScript, Python, Swift, SQL

AI & Agents: Agent workflows, persistent memory, internal agent skills, MCP integrations, sandboxed execution

Backend & Infrastructure: Node.js, tRPC, PostgreSQL, Cloudflare Workers, AWS, Docker, GitHub Actions

Frontend & Desktop: React, Next.js, Tailwind CSS, SwiftUI, macOS